RTCA, Inc.

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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Appendix A

Environmental Test Identification

Important Notice

Information pertinent to this section is contained in Sections 1, 2 and 3. This appendix is applicable for identifying the environmental tests performed.

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Prepared by: RTCA SC-135

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A.1 Introduction and Scope

A need exists to provide a permanent record of the particular environmental test categories that equipment has passed. This need includes post-incident or accident investigation, installation certification, repair, etc. This procedure provides for a paper record (hereafter referred to as the Environmental Qualification Form) to be included in the equipment data package submitted for Technical Standard Order (TSO) authorization and in the installation and maintenance instructions.

Since it is not envisioned that the Environmental Qualification Form will be related to a particular equipment by serial number or date of manufacture, association must be achieved through the equipment type, model or part number. Manufactures should identify the method used to establish traceability to the environmental test categories to which the equipment was tested, including the applicable revision number of the test procedure (Section of RTCA/DO-160) used.

A.2 Environmental Qualification Form (see Figure A-1)

This form provides the necessary information regarding which environmental tests were conducted and, where applicable, the appropriate environmental category of the equipment being tested.

Additional information is included to identify the specific equipment type or model to which the environmental test results apply. A suggested format is depicted in Figure A-1. An example Environmental Qualification Form is shown is Figure A-2 and has been annotated to illustrate a completed form. Equipment manufacturers should expand on the data included on this form to provide added clarity.

In some cases, the manufacturer may wish to qualify the equipment to more than one category for a particular environmental test. In cases where multiple requirements for various categories exist, (e.g., Temperature/Altitude, or Vibration) more than one category may be indicated on the form, but only if the most severe requirement from each of the multiple requirements are met from all of the categories indicated. If all requirements for one category are clearly the most severe, only the most severe category need be identified.

Also, information such as vibration tests conducted with or without shock mounts, fluid tests conducted with Jet A fuel, type of de-icing fluid, and other parameters pertinent to the tests shall be included on the form.

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Figure A-1 Environmental Qualification Form

NOMENCLATURE: _____
TYPE/MODEL/PART NO: _____ TSO NUMBER_____
MANUFACTURER'S SPECIFICATION AND/OR OTHER APPLICABLE SPECIFICATION: _____
_____
MANUFACTURER: _____
ADDRESS: _____
_____
REVISION & CHANGE NUMBER OF DO-160: _____ DATE TESTED: _____

CONDITIONS	SECTION	DESCRIPTION OF TESTS CONDUCTED
Temperature and Altitude	4.0	
Low Temperature	4.5.1	
High Temperature	4.5.2 & 4.5.3	
In-Flight Loss of Cooling	4.5.4	
Altitude	4.6.1	
Decompression	4.6.2	
Overpressure	4.6.3	
Temperature Variation	5.0	
Humidity	6.0	
Operational Shock and Crash Safety	7.0	
Vibration	8.0	
Explosive Atmosphere	9.0	
Waterproofness	10.0	
Fluids Susceptibility	11.0	
Sand and Dust	12.0	
Fungus	13.0	
Salt Fog Test	14.0	
Magnetic Effect	15.0	
Power Input	16.0	
Voltage Spike	17.0	
Audio Frequency Susceptibility	18.0	
Induced Signal Susceptibility	19.0	
Radio Frequency Susceptibility	20.0	

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Figure A-1 Concluded

CONDITIONS	SECTION	DESCRIPTION OF TESTS CONDUCTED
Radio Frequency Emission	21.0	
Lightning Induced Transient Susceptibility	22.0	
Lightning Direct Effects	23.0	
Icing	24.0	
Electrostatic Discharge	25.0	
Fire, Flammability	26.0	
Other Tests		

REMARKS

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- Special Conditions: Include power, special cooling, installation instructions, etc.

Figure A-2 Environmental Qualification Form

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EXAMPLE

NOMENCLATURE: _____
TYPE/MODEL/PART NO: _____ TSO NUMBER _____
MANUFACTURER=S SPECIFICATION AND/OR OTHER APPLICABLE SPECIFICATION: _____

MANUFACTURER: _____
ADDRESS: _____

REVISION & CHANGE NUMBER OF DO-160: _____ DATE TESTED: _____

CONDITIONS	SECTION	DESCRIPTION OF TESTS CONDUCTED
Temperature and Altitude	4.0	Equipment tested to Categories A2, F2
Low Temperature	4.5.1	
High Temperature	4.5.2 & 4.5.3	
In-Flight Loss of Cooling	4.5.4	With auxiliary air cooling, tested to Category W
Altitude	4.6.1	
Decompression	4.6.2	
Overpressure	4.6.3	
Temperature Variation	5.0	Equipment tested to Category B.
Humidity	6.0	Equipment tested to Category A.
Operational Shock and Crash Safety	7.0	Equipment tested to Category B.
Vibration	8.0	Equipment tested to Category S, aircraft zone 4 for fixed wing turbojet engine aircraft, fixed wing unducted turbofan engine aircraft and fixed wing reciprocating/turbojet engine aircraft less than 5,700 kg using vibration test curves W and L.
Explosive Atmosphere	9.0	Equipment identified as Category X, no test performed.
Waterproofness	10.0	Equipment identified as Category X, no test performed.
Fluids Susceptibility	11.0	Equipment identified as Category F Equipment spray tested with phosphate ester-based hydraulic fluid and immersion tested with AEA Type 1 De-icing fluid.
Sand and Dust	12.0	Equipment identified as Category X, no test performed.
Fungus	13.0	Equipment tested to Category F.
Salt Fog Test	14.0	Equipment identified as Category X, no test performed.
Magnetic Effect	15.0	Equipment is Category A.

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Figure A-2 Concluded

CONDITIONS	SECTION	DESCRIPTION OF TESTS CONDUCTED
Power Input	16.0	Equipment tested to Category A(CF), 400 Hz Alternate tests used for AC modulation and inrush AC harmonics and power factor not tested
Voltage Spike	17.0	Equipment tested to Category A.
Audio Frequency Susceptibility	18.0	Equipment tested to Category R(CF).
Induced Signal Susceptibility	19.0	Equipment tested to Category A.
Radio Frequency Susceptibility	20.0	Equipment tested for conducted susceptibility to Category R and for radiated susceptibility to Category R.
Radio Frequency Emission	21.0	Equipment tested to Category H .
Lightning Induced Transient Susceptibility	22.0	Equipment tested to pin test waveform set B, level 3, and cable bundle test waveform set D, Single/Multiple Stroke Level 4, Multiple Burst Level 3.
Lightning Direct Effects	23.0	Equipment identified as Category 2A2A, High Current Strike attachment, High Current Physical Damage
Icing	24.0	Equipment identified as Category X, no test performed.
Electrostatic Discharge	25.0	Equipment tested to Category A.
Fire, Flammability	26.0	Equipment tested to Category B.
Other Tests		Fire resistance tests were conducted in accordance with Federal Aviation Regulations Part 25, Appendix F.

REMARKS

- Tests were conducted at Environmental Laboratories, Inc.
- In the fluids susceptibility tests, material specimens were used
- In the power input test, equipment was tested to subparagraph 16.5.1.4 b, requirement for equipment with digital circuits.

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